

Machine Learning Capstone Project

Objective

The purpose of this project is to apply everything you have learned about **Machine Learning** in a complete real-world scenario.

This project is not about training multiple models and reporting accuracy scores only. Your goal is to understand the problem, investigate and prepare the data, build appropriate Machine Learning models, compare their behavior, evaluate their performance, identify overfitting and underfitting, experiment with hyperparameters, analyze model errors, and communicate meaningful conclusions.

You will also apply **Unsupervised Learning** to investigate hidden structures in the data

Think like a **Machine Learning Practitioner**, not someone who simply executes scikit-learn code.

Group Structure

This is a **group project**.

Each group should consist of **4 students**.

You may divide the implementation tasks among group members. However, every student must understand the complete project and may be asked about **any part of the work during the presentation and Q&A**.

Dataset Selection

Choose a real-world dataset suitable for a complete Machine Learning project.

Your dataset should preferably contain:

- A reasonable number of observations.
- Multiple input features.
- A clear target variable.
- Numerical and/or categorical features.
- Enough complexity to require meaningful preprocessing and analysis.

- A Classification or Regression problem.
- Opportunities for model comparison and interpretation.

Datasets that are extremely small or overly simple are not recommended.

You may choose a public dataset from:

- Kaggle
 - UCI Machine Learning Repository
 - Other reliable public data sources
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Submission Requirements

Submit:

- **Jupyter Notebook / Google Colab Notebook (.ipynb)**
 - **Dataset or dataset source link**
 - **Final presentation**
 - **Short project summary report**
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Project Tasks

Part 1 – Problem and Dataset Understanding

Introduce your Machine Learning problem by answering the following questions:

1. What is the purpose of the dataset?
2. What real-world problem are you trying to solve?
3. What does each row represent?
4. How many rows and columns are there?
5. Which features are numerical?
6. Which features are categorical?
7. What is the target variable?
8. Is the problem **Classification or Regression**?

9. What are your input features X ?
 10. What is your target y ?
 11. Why is Machine Learning appropriate for this problem?
 12. What would a useful prediction mean in the real-world context?
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Part 2 – Initial Data Exploration

Perform an initial inspection of the dataset using appropriate Python and Pandas functions.

Your exploration should include:

- Dataset shape
- Column names
- Data types
- First few records
- Summary statistics
- Target distribution

After exploring the data, answer:

- What did you notice immediately?
- Are there any obvious data quality problems?
- Which variables require further investigation?
- Does the target appear balanced or imbalanced?
- Are any features likely to be irrelevant for prediction?

Do not simply display outputs. Explain what you learned from them.

Part 3 – Data Quality Assessment and Cleaning

Investigate the dataset for data quality problems.

1. Missing Values

Identify:

- Which columns contain missing values?
 - What percentage of each column is missing?
 - What treatment did you choose?
 - Why did you choose this approach?
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2. Duplicate Rows

Investigate:

- Are duplicate rows present?
 - Should they be removed?
 - Could some duplicates be legitimate observations?
 - Explain your decision.
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3. Incorrect Data Types

Check whether any columns are stored using inappropriate data types.

Explain:

- How you identified the problem.
 - How you corrected it.
 - Why the correction was necessary.
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4. Invalid or Unrealistic Values

Search for values that may not be logically valid.

Examples:

- Negative ages
- Impossible measurements
- Invalid dates

- Unrealistic values

For every problem found:

- Explain how you detected it.
 - Explain how you treated it.
 - Justify your decision.
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5. Inconsistent Categories

Check whether categorical values are represented consistently.

Examples:

Male / male / MALE

Riyadh / riyadh / RIYADH

Describe:

- The problem.
 - Your cleaning method.
 - The final result.
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6. Outliers

Detect and investigate outliers using appropriate methods.

You may use techniques such as:

- Box plots
- IQR
- Distribution analysis
- Domain knowledge

Answer:

- Which features contain outliers?

- How many were identified?
- Are they likely to be real observations or data errors?
- Did you keep, remove, transform, or cap them?
- Why?

Do not automatically remove every outlier.

7. Class Imbalance – If Applicable

For Classification problems:

- Is the target balanced?
 - What is the percentage of each class?
 - Is one class significantly less frequent?
 - Why could this affect model evaluation?
 - Which evaluation metrics may be more informative than Accuracy?
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Part 4 – Exploratory Data Analysis

Perform meaningful EDA on the cleaned dataset.

Your analysis should include, when appropriate:

Numerical Analysis

Examples:

- Mean
- Median
- Standard deviation
- Distribution
- Skewness
- Outliers

Categorical Analysis

Examples:

- Frequency tables
- Percentages
- Category comparisons

Relationship Analysis

Investigate important relationships between variables.

Examples:

- Correlation matrix
- Scatter plots
- Grouped summaries
- Box plots by target class
- Pivot tables
- Feature vs Target comparisons

For every important result, explain:

- What did you observe?
- Why might this relationship exist?
- Could this feature be useful for prediction?
- Is the relationship linear, nonlinear, weak, or strong?

Write at least **5 meaningful insights** from your EDA.

Your insights should explain what the data tells you—not simply report numbers.

Part 5 – Feature Preparation

Prepare the features for Machine Learning.

Your work should include the appropriate techniques for your dataset, such as:

- Removing irrelevant features.
- Removing identifier columns when appropriate.

- Encoding categorical variables.
- Scaling numerical features.
- Selecting useful features.
- Preventing Data Leakage.

For every transformation, explain:

- What did you do?
- Why was it necessary?
- Which algorithms are affected by this decision?

For example:

If you apply StandardScaler, explain why feature scaling is important for algorithms such as KNN or SVM.

Do not apply preprocessing techniques without justification.

Part 6 – Data Splitting

Split the data appropriately for model development and evaluation.

Your methodology should include:

- Training data.
- Validation strategy and/or Cross-Validation.
- Final Test Set.

For Classification problems, use stratification when appropriate.

Answer the following questions:

1. What percentage did you use for training and testing?
2. Why did you choose this split?
3. What is the purpose of the Training Set?
4. What is the purpose of the Validation process?
5. What is the purpose of the Test Set?

6. Why should the Test Set remain unseen during model selection?
7. How did you prevent Data Leakage?

Part 7 – Baseline Model

Build a simple baseline model before testing more advanced algorithms.

Examples:

Classification

- Logistic Regression

Regression

- Linear Regression

Evaluate the baseline model and record its performance.

Answer:

- Why did you select this model as your baseline?
- How well does it perform?
- Does it appear to underfit?
- What performance must the more advanced models improve upon?

Part 8 – Machine Learning Model Development

Train and compare **at least four Machine Learning algorithms**.

Choose models that are appropriate for your problem.

Possible Classification Models

- Logistic Regression
- K-Nearest Neighbors
- Decision Tree
- Random Forest
- Gradient Boosting

- Support Vector Machine
- Naive Bayes

Possible Regression Models

- Linear Regression
- Polynomial Regression
- KNN Regressor
- Decision Tree Regressor
- Random Forest Regressor
- Gradient Boosting Regressor
- Support Vector Regression

For **each model**, answer:

1. Why did you choose this algorithm?
2. Is it a linear or nonlinear model?
3. Does it require feature scaling?
4. What does the model learn during fit()?
5. How does it generate predictions?
6. What are its important hyperparameters?
7. What are its main strengths?
8. What are its main limitations?

Do not choose algorithms only to satisfy the required number of models.

Your choices must be justified.

Part 9 – Model Evaluation

Evaluate your models using metrics appropriate for your problem.

Classification

Possible metrics include:

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix
- ROC-AUC when appropriate

Regression

Possible metrics include:

- MAE
- RMSE
- R^2

For your analysis:

- Report the selected metrics.
- Compare the models.
- Explain what each important metric means.
- Explain why the metric is appropriate for your problem.

Do not assume that the model with the highest Accuracy is automatically the best model.

For example, consider whether class imbalance makes Accuracy misleading.

Part 10 – Overfitting and Underfitting Analysis

Investigate whether your models generalize well.

Compare:

Training Performance vs Validation Performance

For your main models, determine whether the model is:

- Underfitting
- Overfitting

- Generalizing well

Explain your conclusion using evidence.

Example:

Training Accuracy: 99%

Validation Accuracy: 81%

Do not only write:

The model is overfitting.

Explain **why the difference between Training and Validation performance indicates overfitting**.

Part 11 – Hyperparameter Experiments

Select important hyperparameters and investigate how changing them affects model behavior.

Examples include:

KNN

- n_neighbors

Decision Tree

- max_depth
- min_samples_split
- min_samples_leaf

Random Forest

- n_estimators
- max_depth

Gradient Boosting

- n_estimators
- learning_rate
- max_depth

SVM

- C
- kernel
- gamma

For each experiment:

- State the parameter being changed.
- Test multiple values.
- Compare Training and Validation performance.
- Explain how the parameter affected model complexity.
- Explain whether it increased or reduced overfitting.
- Select an appropriate value and justify your choice.

Do not simply report the best parameters.

Your goal is to understand **how model behavior changes**.

Part 12 – Model Comparison and Selection

Create a final comparison of your Machine Learning models.

Your comparison should include:

- Model name.
- Training performance.
- Validation performance.
- Final evaluation metric.
- Scaling requirement.
- Main strength.
- Main limitation.

Then select your **Final Model**.

Answer:

1. Which model did you select?
2. Why did you select it?
3. Did it clearly outperform the baseline?
4. Does it generalize well?
5. Is it interpretable?
6. What is its computational complexity?
7. Is it appropriate for the real-world problem?

Do not choose your final model using performance score alone.

Consider:

Performance + Generalization + Interpretability + Computational Cost + Problem Requirements

Part 13 – Error Analysis

Investigate the mistakes made by your final model.

For Classification

Analyze:

- False Positives.
- False Negatives.
- Misclassified observations.
- Classes that are more difficult to predict.

Ask:

- Which type of error occurs most often?
- Which error is more important in the context of your problem?
- Are certain types of observations frequently misclassified?

For Regression

Analyze:

- Residuals.
- Largest prediction errors.
- Target ranges where the model performs poorly.

Ask:

- Does the model systematically overpredict or underpredict?
- Where are the largest errors?
- Is there a visible pattern in the residuals?

Write at least **3 meaningful findings** from your Error Analysis.

Part 14 – Unsupervised Learning: Clustering

Perform an Unsupervised Learning experiment using your dataset.

Do **not** use the Target variable when creating the clusters.

Apply:

- **K-Means**
- At least **ONE additional clustering algorithm**

Choose from:

- Hierarchical Clustering
- DBSCAN
- Gaussian Mixture Model (GMM)

Your analysis should include:

Feature Selection

- Which features did you use?
- Why are these features suitable for defining similarity?

Preprocessing

- Did you need scaling?
- Why?

Model Configuration

For example:

- How did you choose K for K-Means?
- How did you select eps and min_samples for DBSCAN?
- Which linkage method did you use for Hierarchical Clustering?
- How many components did you use in GMM?

Evaluation

Use appropriate methods such as:

- Silhouette Score
- Cluster visualization
- Cluster size analysis
- Cluster profiling

Cluster Interpretation

Do not stop at:

Cluster 0

Cluster 1

Cluster 2

Analyze the characteristics of each cluster.

For example:

Cluster 0:

Higher income and higher spending customers.

Cluster 1:

Lower income and moderate spending customers.

Answer:

- What makes each cluster different?
- Are the clusters meaningful?
- Which variables explain the differences?
- What practical interpretation can be given to each cluster?

Remember:

The algorithm creates the clusters. You are responsible for interpreting what they mean.

Part 15 – Dimensionality Reduction Using PCA

Do not complete this part today.

After the PCA lecture, you will be required to:

1. Prepare the features appropriately for PCA.
2. Apply feature scaling when required.
3. Apply PCA.
4. Examine the Explained Variance Ratio.
5. Determine how many Principal Components should be retained.
6. Compare the original number of dimensions with the reduced dimensions.
7. Visualize the data using Principal Components when appropriate.
8. Train at least one Machine Learning model using PCA-transformed data.
9. Compare model performance before and after PCA.

Answer:

- How many original features did you have?
- How many Principal Components did you retain?
- How much variance was preserved?

- Did model performance improve, decrease, or remain similar?
- Did PCA simplify the dataset?
- Was PCA useful for your particular problem?
- Why?

Important:

PCA does not necessarily improve model accuracy.

Your conclusion should be based on your experiment.

Part 16 – Final Insights and Recommendations

Summarize what you learned from the complete project.

Your conclusion should answer:

1. What were the most important findings from EDA?
2. Which preprocessing decisions had the greatest impact?
3. Which model performed best?
4. Why was it selected?
5. Which models showed overfitting or underfitting?
6. What did you learn from the hyperparameter experiments?
7. What types of errors did the final model make?
8. What did the clustering analysis reveal?
9. What did PCA reveal?
10. What are the main limitations of your solution?
11. What would you improve if you had more time or more data?
12. Would you recommend using the model in a real-world environment? Why or why not?

Your conclusions should explain **what the results mean**, not simply repeat model scores.

Part 17 – Reflection

Answer the following questions:

1. What was the most challenging part of the project?
 2. Which decision required the most critical thinking?
 3. Which algorithm was most difficult to understand or tune?
 4. What result surprised you the most?
 5. What did you learn about model selection?
 6. What did you learn about overfitting and generalization?
 7. If you repeated the project, what would you do differently?
 8. What did you learn about the complete Machine Learning workflow?
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Important Notes

- You will be evaluated on your ability to understand the problem, choose appropriate methods, justify your decisions, evaluate models correctly, and interpret the results.
- Every important preprocessing decision must include an explanation.
- Every model choice must include a justification.
- Every important evaluation result must include an interpretation.
- Do not select the final model based only on the highest score.
- Do not use the Test Set repeatedly during model development.
- Do not apply PCA until the PCA lecture.
- During the presentation, **any group member may be asked about any section of the project.**
- Every student is expected to understand the complete workflow.

Think like a Machine Learning Practitioner whose goal is to transform raw data into a reliable, evaluated, interpretable, and defensible Machine Learning solution.

